www.tntextbooks.in

2.1 Introduction to atom models:

Let us recall the history of the development of atomic models from the previous classes. We know that all things are made of matter. The basic unit that makes up all matter is atom. The word 'atom' has been derived from the Greek word '*a-tomio*' meaning non-divisible. Atom was considered as non-divisible until the discovery of sub-atomic particles such as electron, proton and neutron. J. J. Thomson's cathode ray experiment revealed that atoms consist of negatively charged particles called electrons. He proposed that atom is a positively charged sphere in which the electrons are embedded like the seeds in the watermelon. Later, Rutherford's α-ray scattering experiment results proved that Thomson's model was wrong. Rutherford bombarded a thin gold foil with a stream of fast moving α–particles. It was observed that

(i) most of the α–particles passed through the foil

(ii) some of them were deflected through a small angle and

(iii) very few α–particles were reflected back by 180°

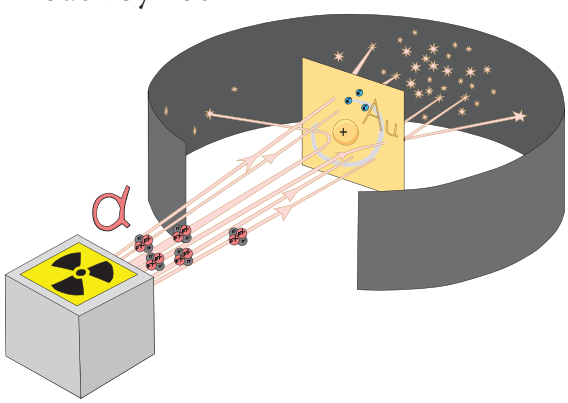


Figure. 2.1 Rutherford's α-ray scattering experiment

Based on these observations, he proposed that in an atom there is a tiny positively charged nucleus and the electrons are moving around the nucleus with high speed. The theory of electromagnetic radiation states that a moving charged particle should continuously loose its energy in the form of radiation. Therefore, the moving electron in an atom should continuously loose its energy and finally collide with nucleus resulting in the collapse of the atom. However, this doesn't happen and the atoms are stable. Moreover, this model does not explain the distribution of electrons around the nucleus and their energies.

2.1.1 Bohr atom model:

The work of Planck and Einstein showed that the energy of electromagnetic radiation is quantised in units of hν (where ν is the frequency of radiation and h is Planck's constant 6.626×10^{-34} Js). Extending Planck's quantum hypothesis to the energies of atoms, Niels Bohr proposed a new atomic model for the hydrogen atom. This model is based on the following assumptions:

1. The energies of electrons are quantised
2. The electron is revolving around the nucleus in a certain fixed circular path called stationary orbit.
3. Electron can revolve only in those orbits in which the angular momentum (mvr) of the electron must be equal to an integral multiple of h/2π.

   i.e. $mvr = nh/2\pi$ -------- (2.1)

where n = 1,2,3,...etc.,